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Design and Implementation of Artificial Neural Network Classifiers based on Hypertuning Parameters for Breast Cancer Diagnosis

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Abstract

Breast cancer is an extremely prevalent disease which impacts women all over the world, and minimizing the death rate among women requires detection at an early stage. This research employs Artificial Neural Network (ANN) to predict and detect breast cancer at an early stage. The suggested model is trained and tested using the 569 patient records taken from Wisconsin Diagnostic dataset for breast cancer. The dataset is split between training and testing sets using 80:20 ratio, with the random states being 0, 1, and 2 for each set. With random state 1 benign and malignant cases are split in same ratio as were in original dataset. The binary cross-entropy function is used to calculate the discrepancy between the predicted class probability and the actual class label. In this research, an Adam optimizer is employed to reduce the loss function. Overall, the parameters of the model are tuned to produce the best outcomes for particular tasks depending on the choice of different random states, activation function, loss function, and optimizer. Several measurements, including training and testing accuracy, confusion matrix, precision, sensitivity, F1_Score, FPR, MCC (Matthews correlation coefficient), AUC, and ROC curve, are employed to evaluate the model performance. Our experimental findings show that the suggested model is capable of making precise predictions and diagnosis of breast cancer.

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Keywords: Breast Cancer; ANN; Binary Cross-entropy Function; Adam Optimizer; Confusion Matrix.

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1. Introduction

Breast cancer is commonly detected and diagnosed using medical imaging procedures including magnetic resonance imaging (MRI), ultrasound and mammography [1][2][3]. However, interpretation of these images is a challenging task, as it requires specialized training and expertise. A feed-forward neural network with numerous layers of linked nodes is known as an ANN. The input features are fed into the network during training using a supervised learning method, and the network's output is then compared to the intended result [4]. To lessen the discrepancy between the output predicted and the output received, the weights of the connections connecting the nodes are changed during training [5]. Once trained, the network may be used to forecast the results given fresh input data.

Deep learning models may be provided to ascertain if a cell is benign or malignant based on its properties, such as form, texture, and size, in the context of diagnosing breast cancer [6][7]. A popular dataset for this purpose is the Breast Cancer Wisconsin diagnostic dataset (BCWD), which includes attributes that indicate properties of the cell nuclei contained in breast tissue samples. Deep learning models may be trained on this dataset to correctly predict the cell's condition, which can help in identification and diagnosis of breast cancer [8].

Deep learning models, like ANN shows potential for early-stage breast cancer prediction and diagnosis accuracy. These models are an essential tool in the campaign against breast cancer since they could enhance patient outcomes and decrease death rates. The survival rate in last five years has grown from 27% to 98% as a result of several screening methods for early identification. This information spurs researchers to improve the effectiveness as well as accuracy of the various breast cancer detection methods.

2. Literature Review

Mammography is a technique to detect the breast cancer by observing the changes in breast tissues [9]. Although certain rules are framed to discriminate malignant and benign but when prediction variables are numerous and interactive, the ad-hoc decision may leads to a false detection and large number of patients need to go for further biopsies.

In mammography equipped with CAD systems, the major problems developed are inconsistency and low classification accuracy. This problem can be overcome with the use of ANN model in CAD and CADx. It motivates researchers to emphasize on the designing of different neural network models with high accuracy and minimum error.

Machine learning approaches, including logistic regression, KNN, SVM, ANN, and random forest was compared by Islam et al. [10] for predicting breast cancer. According to authors, RF and LR models had the lowest accuracy, at 95.7%, while the ANN model had the best accuracy, at 98.57%. The authors also stated that the developed ANN model was more consistent than the other techniques and had the potential to automatically detect breast cancer with high accuracy. A study by A. M. Abdel-Zaher et al. [11] presented a breast cancer classification model based on deep belief networks (DBN).

Using a variety of data mining approaches, Emina and A. Subasi [12] discovered that the rotation forest model had the greatest accuracy of 99.48%. Kalyani Wadkar et al. [13] tested different machine learning methods and author found that ANN performed better than SVM, achieving an accuracy of approximately 97% in testing. They note that ANN has the potential to revolutionize the field of medical diagnosis due to its ability to process data in a more accurate form. In future work, the authors plan to develop an ANN model that can achieve high accuracy without the support of SVM and to reduce processing time by using better algorithms. Sharma Deepti et al. [14] ensemble learning method, which relied on additional trees and neural networks, produced actual accuracy of 99.74%.

3. Methodology

In this research, malignant or benign condition is predicted by ANN model. In proposed research, pre-processing and feature selection of data, splitting of dataset in training and testing with split ratio of 0, 1 and 2, training as well

as optimization of models and performance evaluation using different parameters are incorporated as major building blocks as shown in Figure 1.

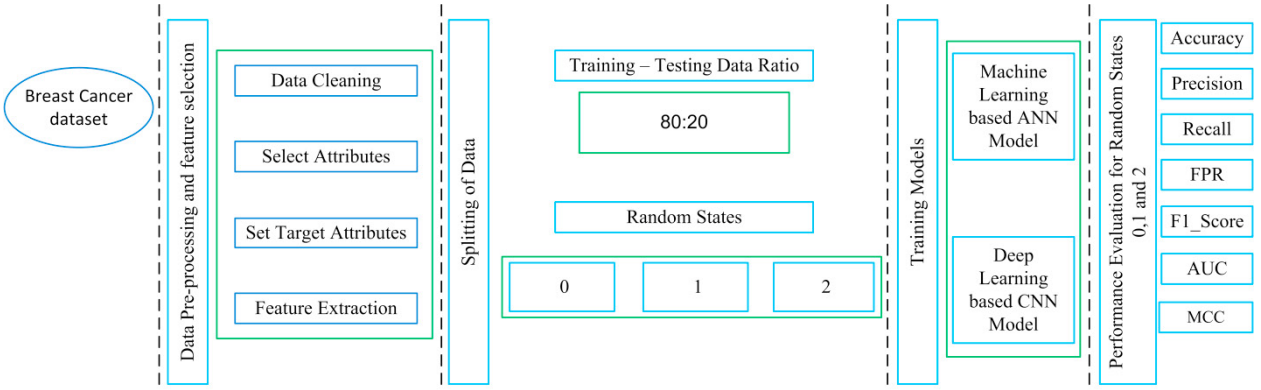


Fig. 1. Block Diagram Representation of Proposed model framework

3.1. Dataset Description

The suggested models are trained and tested using BCWD dataset. The BCWD has 32 columns, the first of which is an ID number and the second of which is a diagnostic result. The final 30 columns, numbered 3 to 32, display the average, standard deviation, and average of the lowest ten attributes. Fractal dimensions, area, symmetry, texture, concave point, density, perimeter, smoothness, indentation, and Radius are the 10 characteristics. These characteristics correspond to the nucleus's size and shape in the target cancer cell. According to Figure 2, this dataset contains 357 benign breast cancer cases and 212 malignant instances, respectively. Malignant instances are categorized as having a value of 1, whereas benign cases have a value of 0.

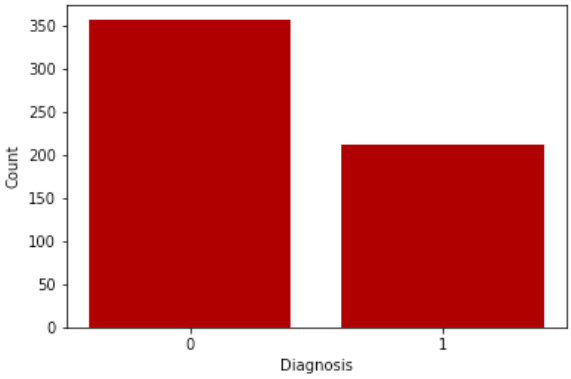


Fig. 2. Number of Benign and Malignant cases in dataset

3.2. Splitting Data Set

With random states 0, 1, and 2, the dataset is split into two portions: training set (80%) and testing set (20%). It is divided into 114 testing data and 455 training data. There are 290 benign and 165 malignant cases in the random state 0 training set, 285 benign and 170 malignant cases in random state 1, and 288 benign and 167 malignant cases in random state 2. With the training set, model is trained and optimized.

4. ANN Model Optimization and Training

To solve the typical regression and classification problems multilayer ANN models are used. The number of neurons present in the input layer of ANN architecture is directly proportional to the number of characteristics in database. The detailed architecture is shown in Figure 3. In this proposed ANN architecture, input layer consists of 30 neurons that are connected with 16 neurons of the first hidden layer as mentioned in Table 1. In this model two hidden layers are used and both hidden layers are connected with 16th -32nd mapped neurons. The detailed parameters used in proposed ANN model are represented in Table 2.

Table 1: Summary Table of ANN Model

Layer (type)	Output Shape	Parameters
dense	(0, 16)	496
dense1	(0, 32)	544
dense2	(0, 1)	33
Total number of parameters: 1,073		
Parameters which are trained: 1,073		
Parameters which are not trained: 0		

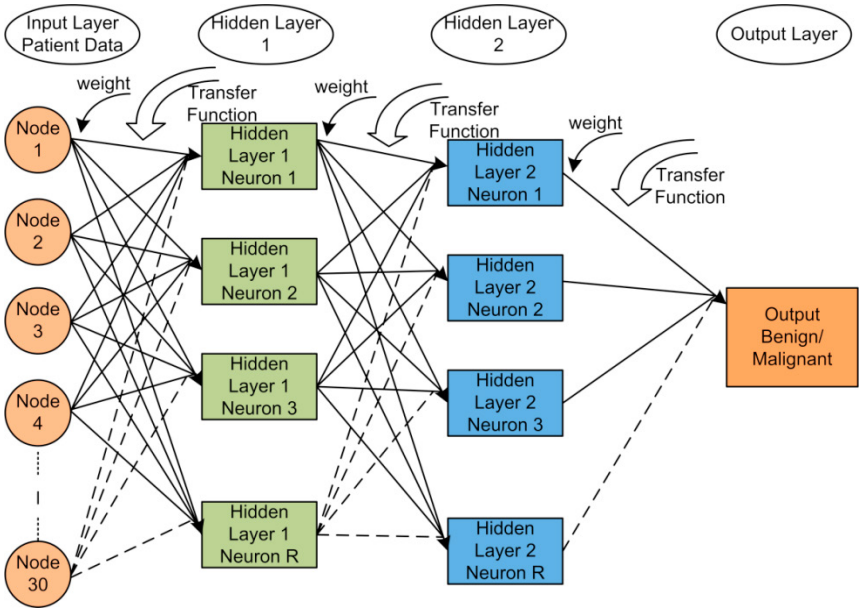


Figure 3. Proposed ANN architecture

Table 2: Parameter used in ANN model

Input node	30
Nodes at Hidden layer_1	16
Nodes at Hidden layer_2	32
Number of epochs	500
Batch size	15
activation function	ReLu, Sigmoid

4.1. Algorithm

Input Layer: A set of input nodes, each of which represents an attribute of the dataset, make up the input layer of ANN. Let X = [x1, x2, ..., xn] be the input vector of size n, where xi is the ith feature.

Hidden Layers: The ANN consists of a set of hidden nodes, where each node represents a neuron that processes the input and passes the output to the next layer. Let H = [h1, h2, ..., hm] be the hidden layer vector of size m, where hj is the jth hidden neuron.

Output Layer: The output layer of the ANN consists of a set of output nodes, where each node represents a class label or a continuous value. Let Y = [y1, y2, ..., yk] be the output vector of size k, where yi is the ith output node.

Forward Propagation: By calculating the weighted sum of the input at each hidden node and then applying an activation function, the input vector X is propagated through the network. Let f(x) be the activation function. The output of the jth hidden node can be represented as follows:

$$h_j = f(\sum_i w_{ji} x_i + b_j) \quad \text{---------(1)}$$

Where b_j is the bias term connected to the jth hidden node and w_{ji} is the weight linking the ith input node to the jth hidden node. Similarly, output of the ith output node can be represented as:

$$y_i = g(\sum_j w_{kj} h_j + b_k) \quad \text{---------(2)}$$

where w_{kj} is the weight between the jth hidden node to the kth output node, and b_k is bias term associated with the kth output node. The function g(x) is the activation function of the output layer.

Backward Propagation: The error is computed using a binary loss function. Let E be the error function. The gradient of error with respect to weight w_{ji} can be represented as:

$$\partial E/\partial w_{ji} = \partial E/\partial y_i \; \partial y_i/\partial h_j \; \partial h_j/\partial w_{ji} \quad \text{---------(3)}$$

Similarly, the gradient of the error with respect to the bias b_j can be represented as:

$$\partial E/\partial b_j = \partial E/\partial y_i \; \partial y_i/\partial h_j \; \partial h_j/\partial b_j \quad \text{----------(4)}$$

Using the gradient descent algorithm, which includes shifting the weights and biases against the gradient at a learning rate α, the weights and biases are updated.

$$w_{ji} = w_{ji} - \alpha \, \partial E/\partial w_{ji} \quad \text{----------(5)}$$

$$b_j = b_j - \alpha \, \partial E/\partial b_j \quad \text{----------(6)}$$

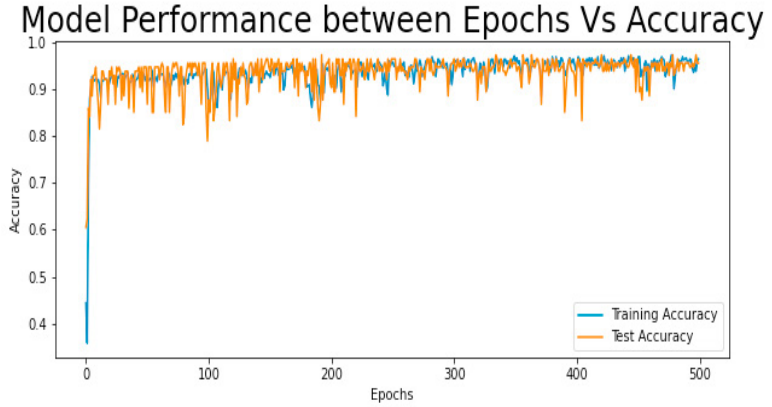
Repeat Steps 4-5: For a predetermined number of epochs or until the error converges to a minimal value, steps 4 and 5 are repeated.

Output: The model is used to forecast the output for fresh input data once training is finished. The input is propagated through the network using the learned weights and biases, and the predicted output is obtained.

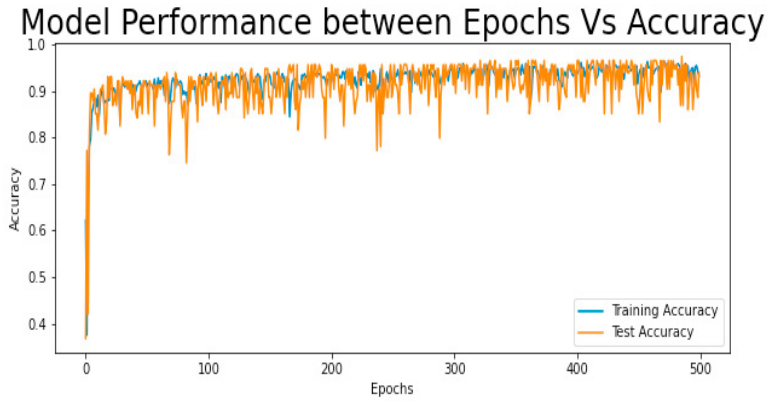
5. Results and Discussion

The performance of ANN model is evaluated and analyzed based on several performance parameters, such as the f1-score, accuracy, recall (sensitivity), precision, confusion matrix, FPR, MCC (Matthews correlation coefficient), AUC, and ROC curve.

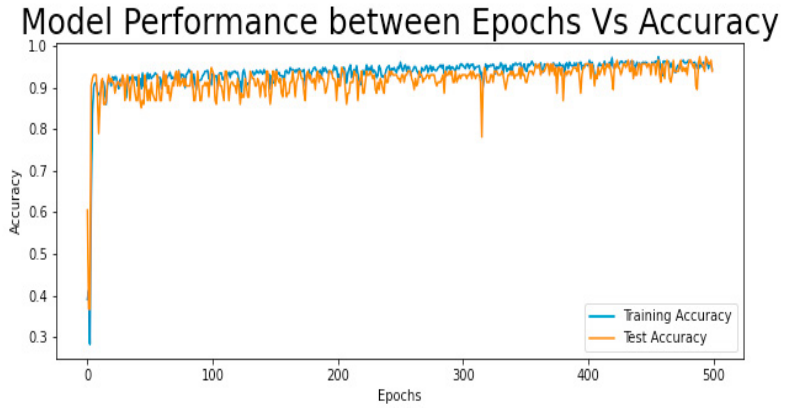
Training and testing accuracy during 500 epochs for ANN model with different random states are shown in given Figure 4. The accuracy of the model improves with the number of epochs.



(a)



(b)



(c)

Figure 4: Graphical representation of Training-Testing Accuracy for ANN [(a,b,c)] models for random states 0, 1 and 2 respectively.

Table 3 provides accuracy percentages for the Wisconsin Breast Cancer Diagnostic dataset for ANN model using different random states. For the ANN model, the highest training and testing accuracy percentages were achieved with random state 1, which were 95.89% and 95.61%, respectively. With random states 0 and 2, where the accuracy percentages were 95.23%/94.73% and 92.34%/91.22%, respectively.

These results imply that the model's performance may be influenced by the choice of random state. Therefore, it is essential to try different random states when training machine learning models to find the optimal configuration that achieves the highest accuracy.

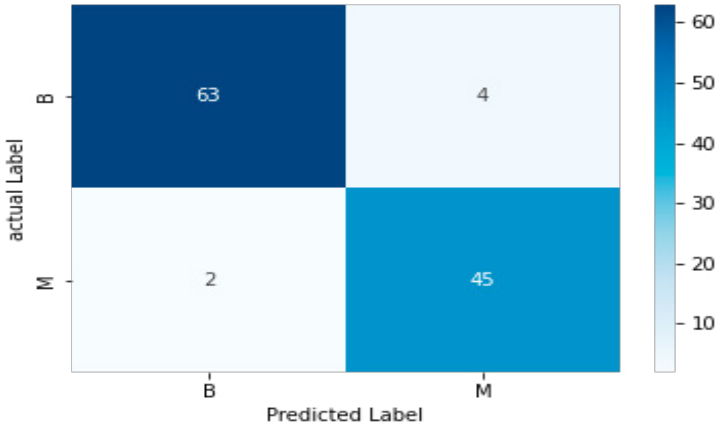
Table 3: Training-Testing Accuracy obtained from proposed models

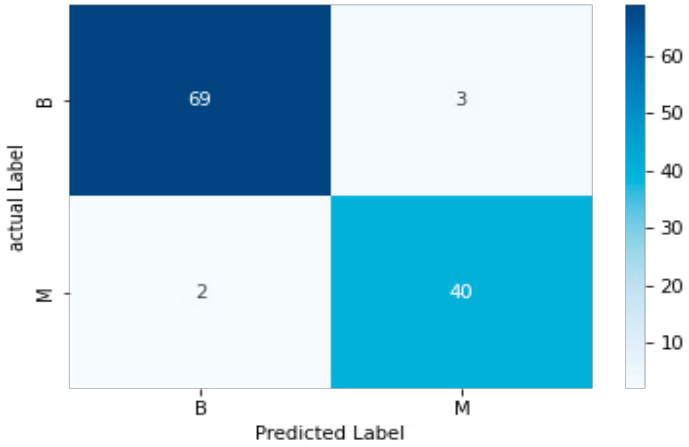
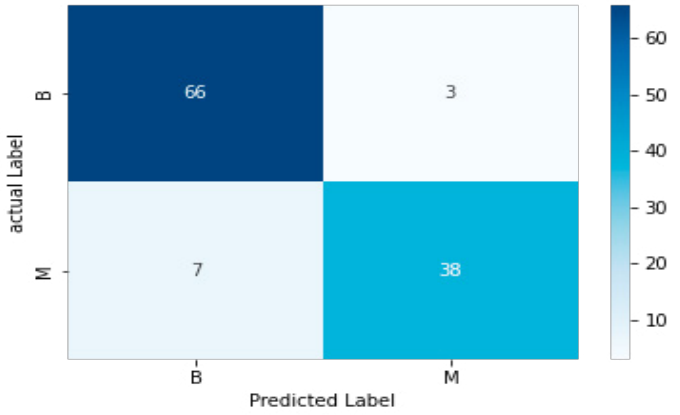
Model	Random State	Accuracy Training set (%)	Accuracy Testing set (%)
ANN	0	95.23	94.73
	1	95.89	95.61
	2	92.34	91.22

The effectiveness of a classification model may be assessed with the use of a confusion matrix. The expected and actual class labels for a piece of data are compiled in a contingency table. The four elements that make up the confusion matrix are True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN). To evaluate the performance of ANN and CNN models on WBCD, a confusion matrix can be constructed for each random state. This can help to determine the sensitivity, accuracy and other performance metrics of the models. The combined confusion matrix is shown in Table 4.

Based on results from the confusion matrix, Table 6 illustrates the classification models' performance metrics for three different random states. For both benign and malignant classifications, these metrics include accuracy, sensitivity, F1 score, FPR, Precision and MCC.

Table 4: Confusion matrix representation for proposed ANN model for Random States 0, 1 and 2

Random State	ANN
0	

1	
2	

The confusion matrix for the ANN model with random state 1 reveals that 109 of the 114 test cases were accurately predicted, including 40 malignant cases that were indeed cancerous and 69 benign instances that were in fact benign. However, 3 instances of the benign class were misdiagnosed as malignant, while 2 cases of the malignant class were misdiagnosed as benign, total 5 cases with inaccurate predictions. The accuracy of the ANN model's random state 1 is greater than that of its random states 0 and 2 as shown in Table 3 and also the proposed classifiers performance for WBCD is shown in table 5.

Table 5: Proposed Classifiers performance Indices for WBCD

Model	Random State	Precision	Sensitivity	F1_score	FPR	Class
ANN	0	.96	.94	.95	.05	B
		.91	.95	.94	.04	M
	1	.97	.95	.95	.04	B
		.93	.95	.93	.04	M
	2	.90	.95	.92	.04	B
		.92	.84	.87	.15	M

ANN model with various random states are used to predict breast cancer using performance assessment criteria. Based on the results in the tables, we can see that ANN model with random state 1 consistently outperform the other random states. The comparative analysis of proposed classifiers with other state of are technologies used by researchers are represented in Table 7.

Table 6: MCC and AUC for proposed classifiers

	ANN		
Random State	0	1	2
MCC	.89	.90	.81
AUC	.98	.99	.97

Table 7: Comparative analysis of proposed classifiers with state of art techniques

Author	Technique	Accuracy	AUC
Varsha Nemade, Vishal Fegade [15]	XGBoost	97	.99
Mohammed Amine Naji, Sanaa El Filali, et. al. [16]	SVM	97.2	.96
Hager Saleh, Sara F Abd-El Ghany, et. al [17]	RF	97.01	-
Emina Aličković & Abdulhamit Subasi [12]	ANN	96.05	.95
Proposed Approach	CNN (RS=1)	97.36	.99

The ROC curve and AUC are important measures of classifier performance, particularly in binary classification problems like breast cancer diagnosis. The ROC curve compares the true positive rate to the false positive rate at different threshold settings to show how well the classifier can discriminate between the two classes. The AUC, a numerical measure of the ROC curve, indicates the probability that a randomly chosen positive example will be scored higher than a randomly chosen negative example by the classifier. AUC values, as shown in Table 6, that are nearer 1 often suggest greater classification performance.

On the basis of results, we can see that ANN model with random state 1 have higher AUC scores compared to the other random states, indicating better classifier performance. For example, the AUC score for ANN with random state 1 is 0.99, which is higher than the AUC scores for ANN with random states 0 and 2.

6. Conclusion and Future Scope

In this proposed research, neural network based ANN classifier is designed and implemented to predict the benign and malignant status of breast cancer dataset perceived from WBCD dataset. The proposed models are trained and tested with the ratio of 80:20 with random states 0, 1 and 2. The performance of artificial neural network classifier for three random states is evaluated and compared. From the results it is concluded that random state 1 provides higher training/testing accuracy than random states 0 and 2 to ANN classifiers as 95.89%/95.61%. As future scope, researchers can apply proposed networks and other machine learning algorithms using new parameters on larger data sets with more disease classes to obtain higher accuracy.

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